PROJECT PROGRESS REPORT

Project: SĀVI - Smart AI Vital Interfacing

Date: March 9, 2026

---

1. COMPLETED MODULES (100% DONE)

Module 1: Mobile Ecosystem & Logic (The SĀVI App)

- **Current Status: Completed & Verified**
- **Details: Developed a fully functional mobile application with a complete navigation stack.**
    - Integrated the **Real-time Telemetry Dashboard** designed to visualize EEG/EMG data.
    - Built the **Emergency SOS Alert System**—this includes the automated logic for caregiver notification and location sharing.
    - **Testing:** The app has been stress-tested using **simulated EEG data packets** to ensure the detection algorithms trigger the UI alerts without latency.

Module 2: Web Presence & Digital Twin (The Website)

- **Current Status: Completed & Deployed**
- **Details:**
    - Official project website built and hosted on **Netlify**.
    - Integrated a **3D Interactive Exhibit** to showcase the device's structural housing.
    - Serves as the central repository for project documentation and remote monitoring links.

2. TECHNICAL INFRASTRUCTURE (80% DONE)

Module 3: Core Firmware & Signal Processing

- **Current Status: Logic Completed / Hardware in Integration**
- **Technical Details:**
  - **ESP32-S3 Configuration:** Successfully programmed the dual-core MCU for high-speed data relay.
  - **SPI Protocol:** Completed the communication bridge between the **ESP32-S3 and ADS1292R** (Analog Front-End).
  - **Digital Filtering:** Written and verified the C++ code for **Notch Filters (50Hz/60Hz)** to remove environmental noise from the brainwave signal.
  - **Note:** The "Brain" of the hardware is fully programmed; we are currently awaiting the final assembly of the re-optimized electrode headband for field testing.

3. WORK N PROGRESS (PHASE 2)

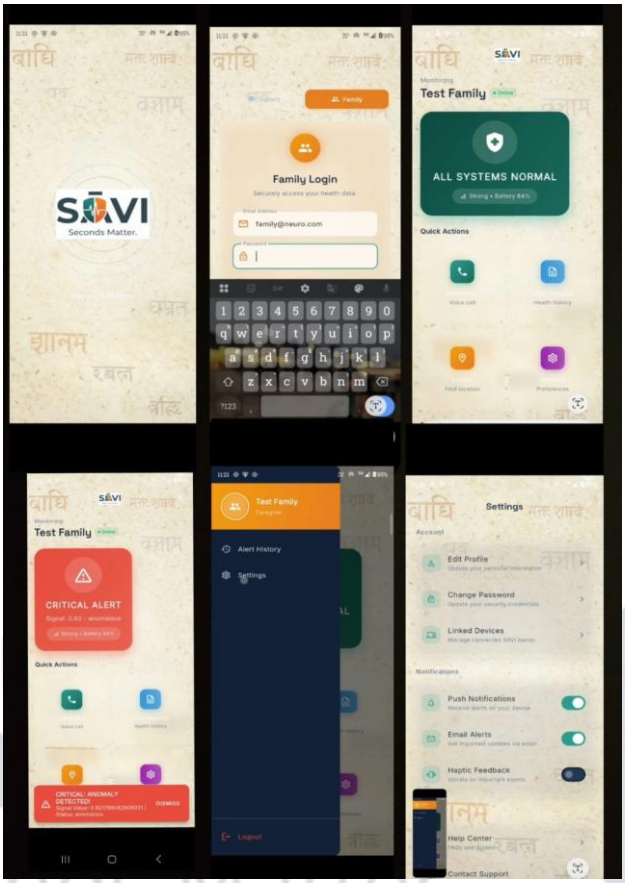
- **Hardware Housing:** 3D printing the final casing for the ESP32 and battery module.
- **Final Calibration:** Tuning the detection thresholds for real-human EEG signals once the high-impedance headband arrives this week.

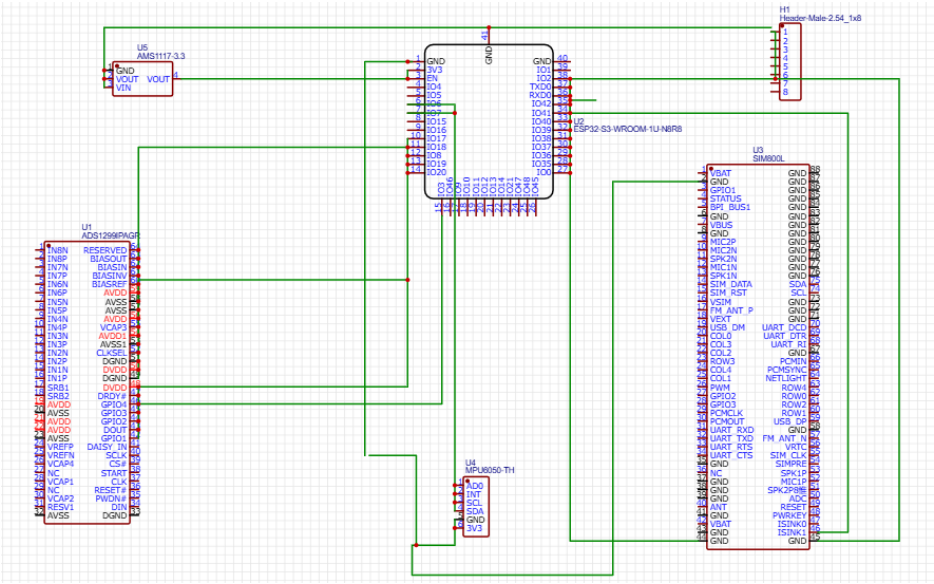
SāVI
Seconds Matter

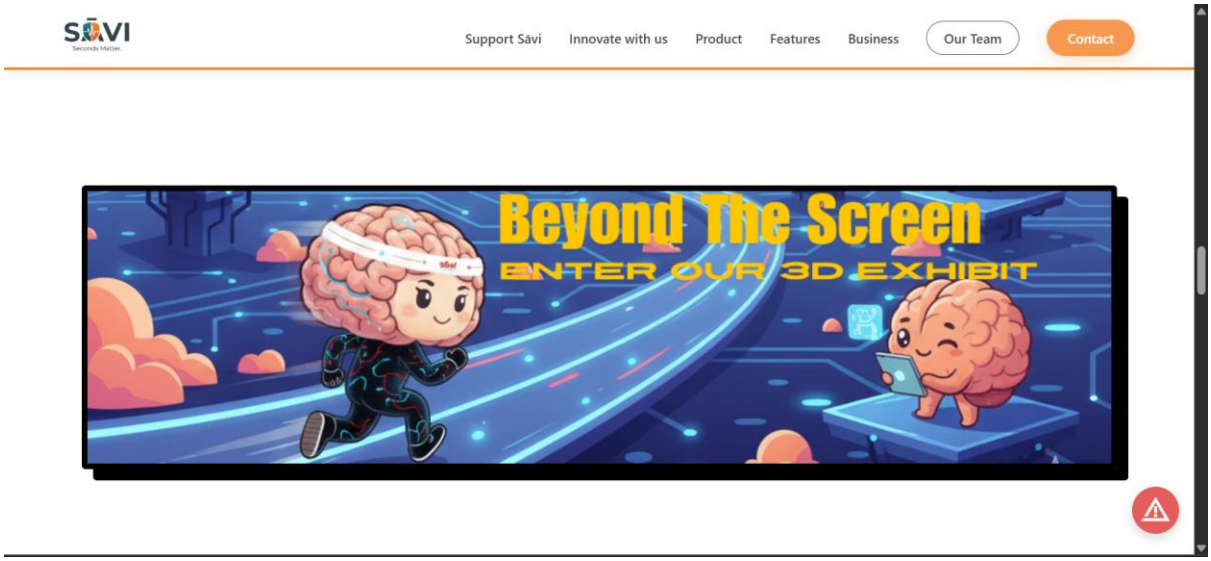
4. EVIDENCE OF PROGRESS

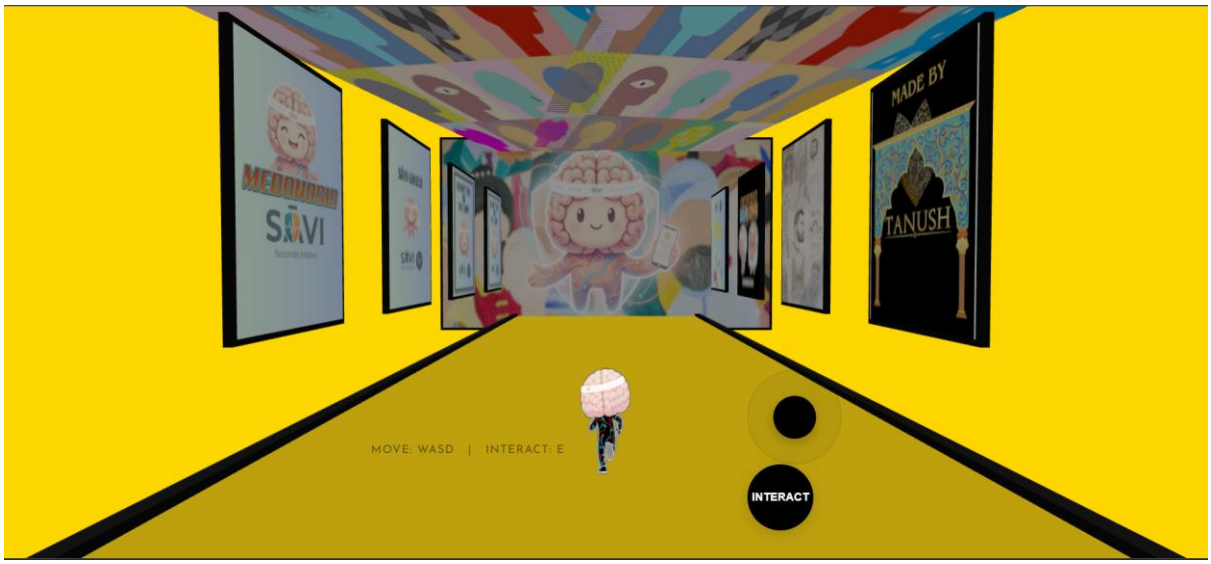
Evidence Type	Description
YouTube Demo	https://youtube.com/shorts/BBRr0LT1sO0?si=M98-NdQfLFRxCOB4
GitHub Repo	https://github.com/tanush69lucifer/Saavi/tree/main/hardware
Netlify Link	https://savi-lab-studio.netlify.app/

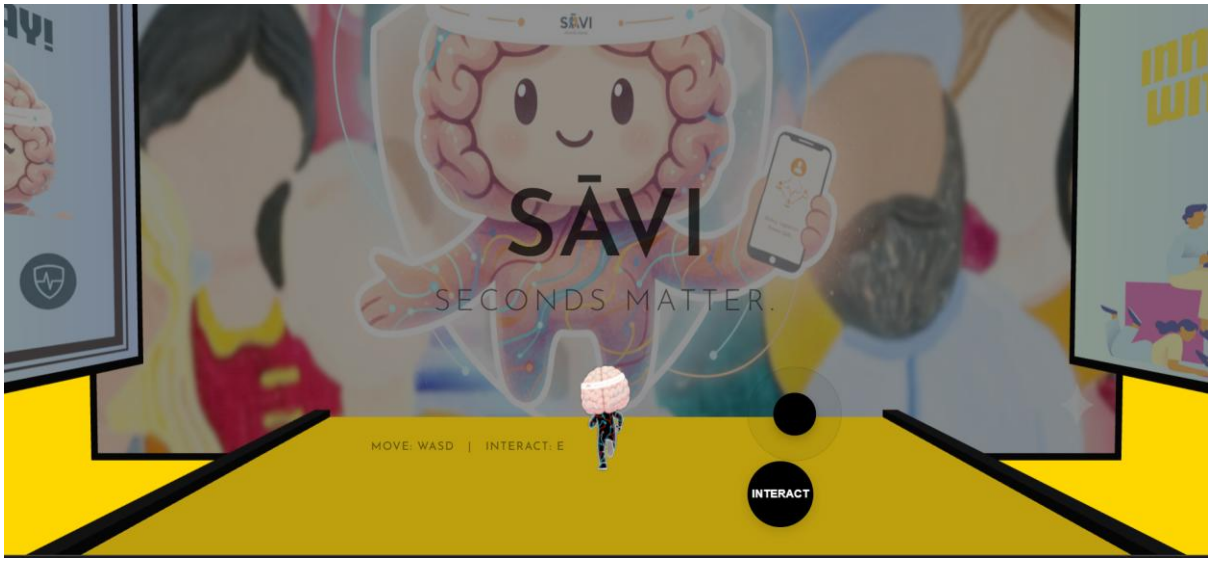
5. PROOF OF EVIDENCES






SĀVI
Support Sāvi
Innovate with us
Product
Features
Business
Our Team
Contact
Beyond The Screen
ENTER OUR 3D EXHIBIT


MEDINDSIA
SĀVI
MADE BY
TANUSH
MOVE: WASD | INTERACT: E
INTERACT


SĀVI
SECONDS MATTER.
MOVE: WASD | INTERACT: E
INTERACT